Preferences, Alignment, Fatigue, and Burnout: A Cross-Sectional Shift-Schedule Survey of Emergency Department Nurses and Attending Physicians

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Background

49.8%[1] of emergency physicians report burnout, the highest of any specialty

31%[2] of emergency nurses report high emotional exhaustion

2.5×[3] higher likelihood of burnout among nurses working ≥ 10-h shifts

- The emergency department (ED) operates around the clock, and burnout is more common among ED clinicians than in most other professions. [1,2]
- Shift scheduling is one contributor to fatigue and burnout that health systems can change; for example, longer shifts are linked to more burnout among nurses. [3]
- Little is known about how ED nurses and attending physicians differ in their scheduling preferences and recovery needs, or whether working a schedule that matches those preferences is linked to lower fatigue and burnout.

Objectives

1. Compare the **scheduling preferences** of ED nurses and attending physicians and the **reasons** underlying them.
2. Compare when they **start feeling fatigued** during a shift, and how much time off clinicians need to **recover** after one shift and after several shifts in a row.
3. Examine whether **schedule alignment** (the degree to which a clinician's most recent schedule matched their preferences) is associated with **burnout and fatigue**.

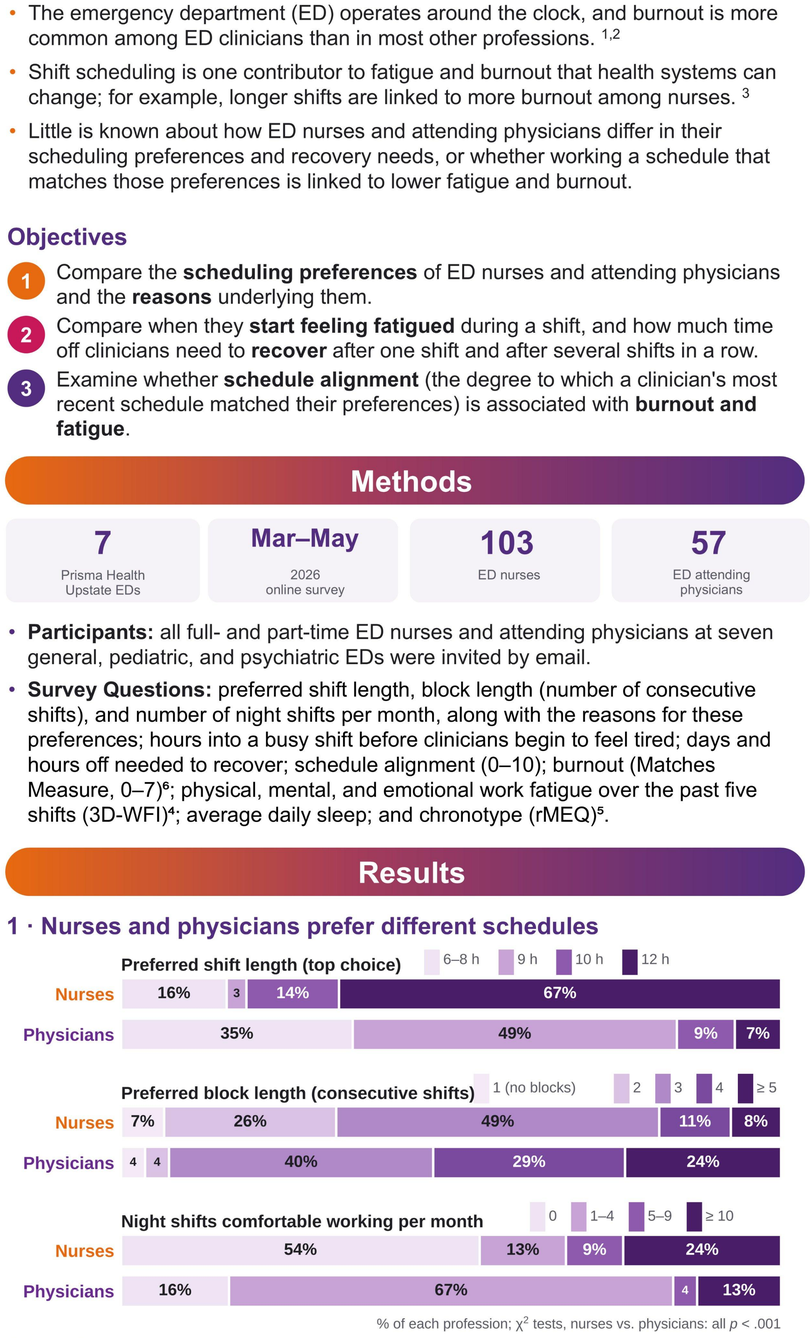
Methods

7	Mar–May	103	57
Prisma Health Upstate EDs	2026 online survey	ED nurses	ED attending physicians

- **Participants:** all full- and part-time ED nurses and attending physicians at seven general, pediatric, and psychiatric EDs were invited by email.
- **Survey Questions:** preferred shift length, block length (number of consecutive shifts), and number of night shifts per month, along with the reasons for these preferences; hours into a busy shift before clinicians begin to feel tired; days and hours off needed to recover; schedule alignment (0–10); burnout (Matches Measure, 0–7)[6]; physical, mental, and emotional work fatigue over the past five shifts (3D-WFI)[4]; average daily sleep; and chronotype (rMEQ)[5].

Results

1 · Nurses and physicians prefer different schedules



% of each profession; χ^2 tests, nurses vs. physicians: all $p < .001$

Figure 1. Preferred shift length, block length, and night shifts per month.

2 · What shapes clinicians' scheduling preferences

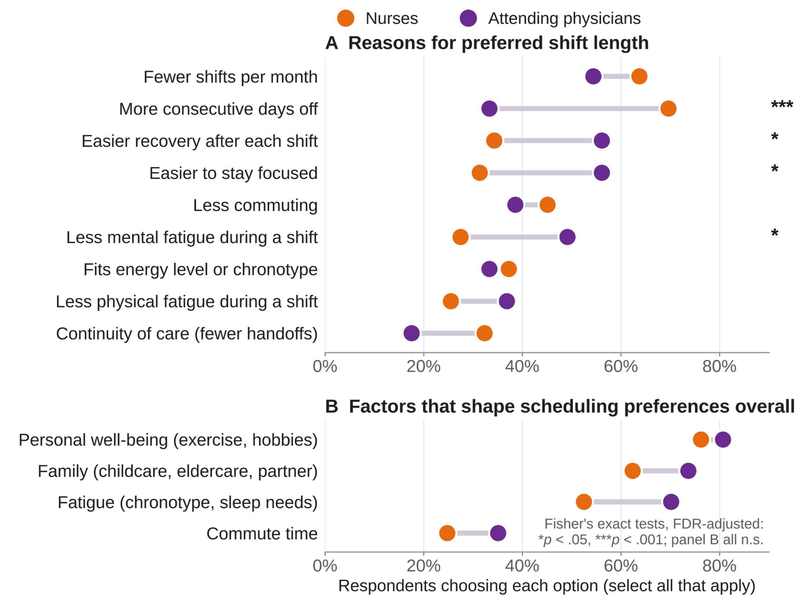


Figure 2. Reasons for the preferred shift length in Figure 1 (A) and factors that shape scheduling preferences overall (B).

3 · Preferred shifts outlast fatigue onset

76% of clinicians prefer shifts that last **beyond the point where they begin to feel fatigued** (median 2 h longer).

Hours into a busy shift before fatigue begins

	Day	Evening	Night	Most common preferred shift
Nurses	8.4 h	7.6 h	6.9 h	12 h
Physicians	6.9 h	6.4 h	6.0 h	9 h

Means; earliest on night shifts in both groups (Friedman tests, $p < .001$)

4 · Clinicians need more time off after more shifts in a row and after night shifts

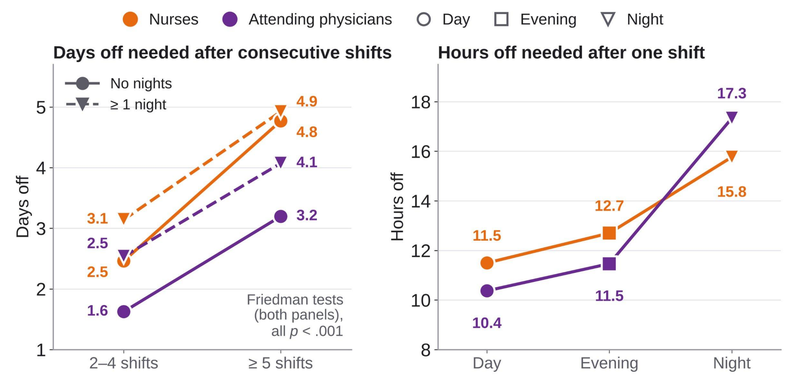


Figure 3. Mean days off needed after 2–4 or ≥ 5 shifts in a row, and hours off needed after one shift.

5 · Nurses rated their schedules a better fit

- Nurses rated their most recent schedule as a **better match to their preferences** than physicians did (**6.8 vs. 5.3** of 10, $p < .001$).
- Nurses reported **more physical fatigue** (**3.1 vs. 2.1** of 5, $p < .001$), **mental fatigue** (3.1 vs. 2.6, $p = .044$), and **overall fatigue** (3.0 vs. 2.4, $p = .007$); emotional fatigue did not differ ($p = .38$).
- **Burnout did not differ** (**3.3 vs. 3.3** of 7, $p = .97$), and average sleep was similar (6.7 vs. 6.9 h, $p = .42$).
- Because the professions differed in schedule fit and fatigue, **Figure 4 adjusts for profession**.

6 · Better schedule fit, less burnout and fatigue

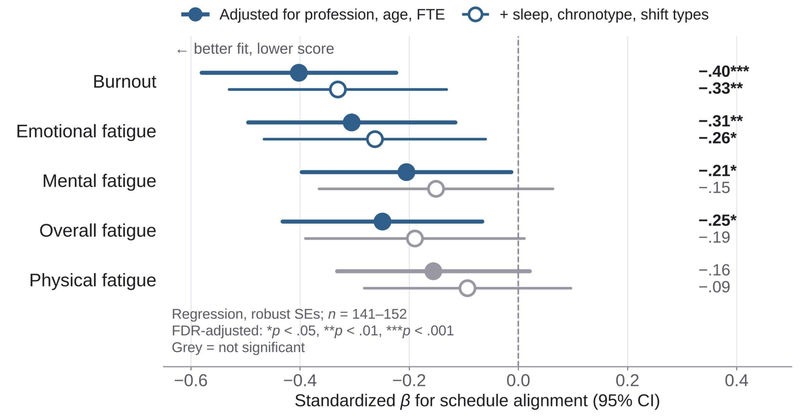


Figure 4. Association of schedule alignment with each outcome (standardized β). The association with burnout was significant among both nurses and physicians, and alignment explained an additional 9% of the variance in burnout beyond profession, age, FTE, sleep, and chronotype ($p < .001$). Longer sleep was also associated with lower burnout ($\beta = -.25$, $p = .007$).

Discussion

- ED nurses and attending physicians differed in their **preferred schedule structures**: nurses most often preferred **12-h shifts**, whereas physicians most often preferred **9-h shifts**.
- **Most clinicians (76%)** preferred shifts **longer than the point at which they begin to feel tired** during a shift. When choosing schedules, both professions cited **personal well-being and family needs** at least as often as fatigue.
- Clinicians reported needing **more time off after more consecutive shifts and after night shifts**. Given evidence linking nurse shifts of 10 h or longer to burnout,[3] schedules that include longer shifts may also need to **protect recovery time**.
- **Greater schedule alignment** was associated with **lower burnout** and lower emotional, mental, and overall fatigue; associations with burnout and emotional fatigue **persisted after adjustment** for sleep, chronotype, and shift type. **Longer sleep** was independently associated with **lower burnout**.
- Eliciting and accommodating clinicians' **scheduling preferences**, while preserving adequate recovery time, **may help reduce burnout**. Given the cross-sectional, self-reported design, **prospective studies** should test these associations and examine whether accepting in-shift fatigue in exchange for more days off reduces longer-term fatigue.